

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 7.3b: LLM Limitations and Transition to RAG

Zhigang Feng

Spring 2026

Topic 7.3b Agenda

1. **Three structural limits** — knowledge cutoff, hallucinations, context window
2. **Lost-in-the-middle & prompt sensitivity** — positional saliency, paraphrase robustness
3. **Grounded generation preview** — cite-then-claim contract; bridge to Lec08/Lec09
4. **LLMs as measurement instruments** — AI-exposure scores from Felten/Webb to Eloundou
5. **Replication: Yin–Vu–Persico (NBER 35110, 2026)** — when only the LLM changes, regression coefficients can flip
6. **Transition to RAG** — why retrieval-augmented generation

Arc: T1a Why+Tokens → T1b Embeddings → T2a Attention → T2b Transformer → T3a Training → T3b Limits → T4 Applications.

Limitations

The Knowledge Cutoff

- Every pretrained LLM is trained on a *static snapshot* of text. After the snapshot ends, the model learns nothing new.
- Each model has a **knowledge cutoff date**. After this date:
 - Events are invisible to the model
 - New papers, datasets, policy announcements simply don't exist as far as it is concerned
 - Asking “what happened in March 2026?” against a model trained through 2024 will at best yield “I don't know” — and at worst, a confident fabrication
- **Why not just retrain?**
 - Cost: \$10M–\$100M+ for a frontier model
 - Time: weeks to months
 - Energy and infrastructure constraints
 - Practically impossible to keep “current”
- This is not a bug — it is structural. Any solution that needs *fresh* information must work *around* the cutoff, not against it. **This is the core motivation for Lecture 8.**

Hallucination Risk

What it looks like

- Fabricated citations to papers that do not exist
- Made-up statistics that look plausible
- Non-existent laws, cases, article numbers
- Subtly wrong details in otherwise correct prose

Mitigations

- Require and verify citations externally
- Use retrieval (Lec08) to ground answers in real documents
- Use tools / code: numbers from a calculator
- Validate a sample by hand
- Set temperature $\tau = 0$ when measuring

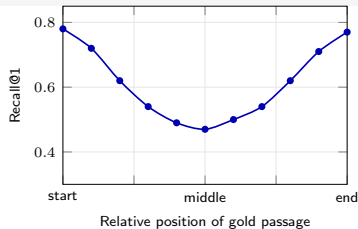
Why does it happen? The model is trained to produce *plausible-sounding* next tokens, not *true* ones. Truth is a property of the world; plausibility is a property of language.

Context Window Limits

- Even modern models with 100,000+ token context cannot swallow arbitrarily large datasets:
 - A 200-page 10-K filing easily exceeds short context windows
 - Decades of FOMC transcripts cannot be loaded at once
 - Multi-firm or multi-country panels of textual data certainly cannot
- **Practical consequences:**
 - Long inputs must be *chunked*, which can break coherence and cross-reference
 - Important information may sit outside the window altogether
 - Attention cost is $O(n^2)$ in the context length, so doubling n quadruples compute and memory
- Newer architectures (sparse attention, sliding-window attention, RoPE-extension, state-space hybrids) push the limit but do not eliminate it.
- **Implication:** you cannot just dump “all the data” into a prompt. You need a way to *retrieve* the relevant pieces and feed them in — another argument for RAG.

Lost in the Middle: Positional Saliency in Long Contexts

- Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni, Liang (TACL 2024): when the relevant fact sits in the *middle* of a long prompt, recall drops 20–30 percentage points relative to its position at the start or end.
- **U-shaped recall curve.** Across closed (GPT-3.5/4) and open (LLaMA-2, MPT) LLMs, accuracy is high when the gold passage is near the start or the end of the context window and sags in the middle — a positional bias in attention, not random noise.
- **Two implications:**
 - *Stuffing the prompt* (concatenate as much as it fits) is **not a substitute for retrieval**: padding adds noise that buries the signal.
 - RAG (Lec08) must therefore *rank* chunks, not merely retrieve them. Top-ranked chunks should be placed at the prompt's ends.



Stylized U-shape from Liu et al. (2024); same qualitative pattern across model families.

Prompt Sensitivity — The Same Sentence, Two Labels

Generic prompt:

Classify this sentence:

"Inflation remains elevated and the Committee remains highly attentive to inflation risks."

Label:

Model output: *neutral* — the descriptive verbs ("remains") and the absence of a directional action word dominate.

Role-conditioned prompt:

You are a monetary economist. Is the following sentence hawkish, dovish, or neutral?

"Inflation remains elevated and the Committee remains highly attentive to inflation risks."

Label:

Model output: *hawkish* — "highly attentive to inflation risks" is read as a tightening signal once the prior is monetary policy.

- Both prompts pass the same sentence to the same model. The difference is the *prior* the role + label space imposes (cf. ICL as Bayesian inference, T3a).
- **Brucks & Toubia (2025, PLOS ONE)**: primacy bias on balanced binary tasks — GPT-4 frequently picks the *first* option (a strong, robust primacy bias). Randomize option order; average across runs.
- **WEIRD bias & version drift**: training corpus is Western/English-leaning; closed-source models change silently between API calls.
- **Take-away**. Treat the prompt as part of the research design — freeze it, version it, report it. Robust analyses report results under multiple paraphrases.

LLMs as Measurement Instruments

The Literature's Question: How Exposed Is Each Job to AI?

- **The measurement problem:** map $\sim 19,000$ O*NET tasks (or ~ 900 SOC occupations) to a numeric *AI exposure* score that can be regressed on wages, employment, and county outcomes.
- **Two pre-LLM generations of measures:**
 1. **Felten, Raj, Seamans (2018, 2021)** — AI Occupational Exposure index (AIOE). Link AI benchmark progress (image recognition, translation, ...) to O*NET ability requirements. *Mechanical mapping; human raters set the abilities.*
 2. **Webb (2020)** — text overlap of AI patent abstracts with O*NET task descriptions. *No raters at all — pure TF-IDF.*
- Both treat the measurement instrument as **fixed and inspectable**. A re-run next year produces the same numbers.
- **Economist's framing:** exposure is a *measurement-error* question, not a CS question. The next slide shows what changes when the instrument is itself an LLM.

Three Generations of AI-Exposure Instruments — Compared

Same target (occupation-level exposure score), three very different instruments:

	Felten, Raj, Seamans (2018, 2021)	Webb (2020)	Eloundou et al. (2023)
Instrument	Hand-curated map: AI benchmark → O*NET ability	TF-IDF cosine: AI patent abstracts → O*NET tasks	Single LLM call: GPT-4 rates each task on 4-level rubric
Cost per refresh	High (human raters, weeks)	Low (re-tokenize, rerun)	Low per call; high per provider switch
Stability across re-runs	Identical (deterministic mapping)	Identical (deterministic)	Stochastic ; different provider = different score
Validity assumption	Benchmark progress proxies real capability	AI patent text proxies AI capability	The LLM is a calibrated rater
Auditable by economist?	Yes (rater notes)	Yes (read the patent and the task)	Partially (rationale text); not the model weights
Failure mode	Benchmarks lag the frontier	Patent corpus is noisy	Version drift, hallucination, prompt sensitivity

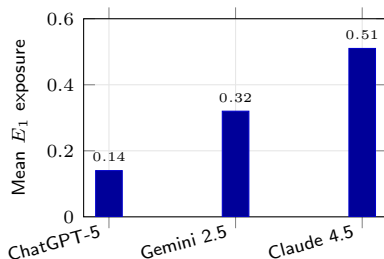
The LLM-as-Rater Turn: Eloundou et al. (2023)

- **Eloundou, Manning, Mishkin, Rock (2023, “GPTs are GPTs”)** — the first paper to use **GPT-4 itself** as the rater on the O*NET task list.
- **Rubric (per task):**
 - **E0** — no LLM speedup.
 - **E1** — direct LLM access reduces task time by $\geq 50\%$.
 - **E2** — speedup achievable with LLM-powered software built atop the model.
- **Procedure:** prompt GPT-4 with a task description, ask it to classify, repeat for $\sim 19\text{k}$ tasks, average within occupation $\rightarrow$ exposure score.
- Quickly adopted as the *default* exposure measure — Acemoglu (2024), IMF (2024), BIS, and many follow-ups all rebuild on this LLM-rated input.
- **The instrument is now the same kind of model as the subject of the research** — and as Topic 7.3a established, that instrument is stochastic, hallucination-prone, and version-drifting.

Replication: Yin, Vu, Persico (NBER 35110, 2026)

Design — only the LLM changes:

- Same Eloundou (2023) prompt and task list (~19k O*NET tasks, ~900 occupations).
- Three frontier 2026 annotators: **ChatGPT-5**, **Claude 4.5**, **Gemini 2.5**.
- Identical decoding (low temperature, deterministic where supported); identical few-shot examples; identical aggregation to occupation-level E_1 means.
- Plug the resulting exposure scores into the same downstream difference-in-differences regressions used by the labor-AI literature.



Same prompt, same tasks — $3.6\times$ divergence in mean exposure.

- Pairwise agreement: **56.9%**, Cohen's κ as low as **0.36**.
- Individual DiD coefficient varies **$2.4\times$** across annotators.
- County-level estimate **flips sign** (significant negative $\rightarrow$ insignificant positive).

What 0.36 means (Landis–Koch benchmarks): <0.20 slight, $0.21\text{--}0.40$ *fair*, $0.41\text{--}0.60$ moderate, $0.61\text{--}0.80$ substantial, $0.81\text{--}1.00$ almost perfect. Publication-grade hand-coding aims for $\kappa \geq 0.75$ ^{11 / 17}

Attitude in the Literature: LLMs Are Not Static Instruments

- **The error is non-classical:** disagreement concentrates on tasks near the rubric boundary — the high-exposure tasks the regressor cares about. Errors correlate with the regressor.
- Yin et al.: *“the risks of treating evolving LLMs as static instruments.”* A paper rated with GPT-4 in 2023 cannot be re-run in 2026 and get the same labels.
- **Each T3b limitation → a measurement consequence:**
 - *Sampling stochasticity* → set $\tau = 0$ and report it.
 - *Hallucination / mis-calibration* → ensemble across models, report Cohen's κ .
 - *Version drift* → pin the snapshot; treat *prompt + model + seed* as one instrument.
- **Emerging best practice:** pre-register prompt + model + decoding; use multiple frontier annotators and report between-model agreement; audit against a hand-coded gold standard (Topic 7.4); report inter-annotator sensitivity, not a single point estimate.
- **Takeaway:** the LLM is a measurement instrument — calibrate, version, and disclose.

When NOT to Use an LLM Rater — A Decision Rule

Pearson 0.6 from 50 lines of `re.findall` is a real signal. The LLM rater must clear a meaningful bar to justify its cost.

A defensible workflow before reaching for an LLM:

1. **Hand-code 100–200 task gold standard** (you, an RA, two coders for κ).
2. **Fit a simple baseline** — TF-IDF + logistic regression, or a keyword rule with handful of features.
Report κ vs. gold.
3. **Run the LLM in parallel.** Compute $\kappa(\text{LLM}, \text{gold})$ and $\kappa(\text{LLM}, \text{baseline})$.
4. **Decide:**
 - $\kappa(\text{LLM}, \text{gold}) \geq 0.75$ and substantially $> \kappa(\text{baseline}, \text{gold}) \Rightarrow$ LLM is validated; use it, disclose model + prompt.
 - $\kappa(\text{LLM}, \text{gold}) < 0.60$ or $\approx \kappa(\text{baseline}) \Rightarrow$ ship the baseline; LLM doesn't earn the version-drift risk.
 - Between 0.60 and 0.75 $\Rightarrow$ ensemble (LLM + baseline + tie-break by hand on disagreements).

The LLM is one instrument among many, not a default. The Yin–Vu–Persico replication shows what you risk by skipping the audition.

Transition to RAG

Grounded Generation: The Linchpin for Lec08 and Lec09

- **Why hallucinations happen.** A pretrained LLM is a *parametric memory*: facts are baked into the weights, mixed with patterns, and re-emerge probabilistically through next-token sampling. Plausibility is what the loss optimizes; truth is not.
- **Two ways out — both grounded:**
 - **Retrieve** the source text and condition the answer on it (**RAG, Lec08**). The model writes “Brainard (2017) reports...” *after* the relevant Brainard paragraph appears in the prompt.
 - **Call a tool** that returns ground-truth data (**agents, Lec09**). For a numeric fact, the model asks `fred_query(UNRATE, ...)` and reads off the actual series.
- **The contract: “cite, then claim.”** Any factual statement must be paired with the source the model is conditioning on — a retrieved passage, a tool output, an attached file. Without that pairing, the statement is asserted from parametric memory and is auditable only by hand.
- **The big structural picture of the second half of the course:**

$$\underbrace{\text{LLM}}_{\text{language engine (Lec07)}} + \underbrace{\text{retrieval}}_{\text{grounded memory (Lec08)}} + \underbrace{\text{tools}}_{\text{grounded action (Lec09)}} \Rightarrow \text{auditable measurement.}$$

The Brain Metaphor

An LLM is like a very capable brain that finished school on a fixed date.

What the brain knows

- General language and reasoning
- Broad economics, math, programming
- Published papers up to its cutoff
- Common patterns, idioms, code recipes

What the brain does not know

- Anything that happened after the cutoff
- Your private FOMC corpus
- A specific firm's internal documents
- This morning's data release
- Your co-author's unpublished draft

Question: how do we let this brain access the information on the right *without* retraining it?

Next Step: Extending the Brain

- **Three options for adding new knowledge to an LLM:**
 1. **Retrain** from scratch with the new data. Effective, but expensive (\$M–\$B), slow (months), impractical for individual researchers.
 2. **Fine-tune** on the new data. Cheaper, but unreliable for injecting new *facts* — fine-tuning is good at changing style and behavior, less so at memorizing isolated facts.
 3. **Retrieve** relevant pieces at inference time and inject them into the prompt. Cheap, dynamic, and the model never has to be touched.
- **Lecture 8** is dedicated to option 3: **Retrieval-Augmented Generation (RAG)**. A frozen LLM combined with a queryable external store of documents.
- **Lecture 9** goes one step further — **agentic AI**, where the brain not only retrieves information but also *acts*: runs code, fetches URLs, writes files, calls other tools. Claude Code is the running example.
- **Lecture 10** brings it all together with case studies of how economists actually use these tools today.

Summary

1. **Inputs:** tokenization + embeddings + positional encodings turn text into a sequence of vectors.
2. **Self-attention** is the core operation of the Transformer: Q, K, V projections $\rightarrow$ scaled dot-product scores $\rightarrow$ softmax $\rightarrow$ weighted sum of values.
3. **Stacking** layers (residuals, normalization, feed-forward networks) yields contextualized representations.
4. **Generation** is autoregressive: project the last embedding to the vocabulary, sample, append, repeat.
5. **Established applications:** TF-IDF + cosine (Kogan et al.), Word2Vec + BVAR (ECB Word2Prices, Araujo et al.), fine-tuned RoBERTa (CentralBankRoBERTa, Pfeifer & Marohl).
6. **2024–25 frontier:** GPT-4 for Fedspeak (Hansen & Kazinnik); LLM embeddings for cross-sectional returns (Chen, Kelly & Xiu); transformer in SDF (Kelly et al.); asset embeddings from holdings (Gabaix et al.).
7. **LLMs have a fixed knowledge cutoff.** Next: extend the brain with retrieval — **Lecture 8: RAG.**